

premature to write an obituary for C, but if you are in the industry and a master of C alone, then things might get rough for you in the future.

With seven of the top ten languages belonging to the C-family of programming languages, C will forever be the favourite of those in academia as the first programming language taught. The seven programming languages in this list that belong to the C-family are C, C++, C#, Go, Java, PHP and Swift. The three open source compilers of C for the Linux platform are *gcc*, *clang* and *pcc*. C is used in the development of the Linux kernel and open source software like Mozilla Firefox, Wireshark, MySQL, Android, Blender, GIMP, Mozilla Thunderbird, VLC Media Player, VirtualBox, CMake, MariaDB, PyGTK, SciPy, MongoDB, PHP, etc. So, with this history of prolific open source development, we can safely conclude that C has a high Openness Score.

C++

C++ is also a programming language used for developing mobile, enterprise and embedded applications. C++ is a compiled language with imperative and object-oriented features. In the beginning, C++ was just a version of C with built-in support for classes and objects. But over the years, C++ has become mature and independent, becoming widely used in the industry. The ranking of C++ has stabilised over the years, and it regularly features among the top five in most lists. The popularity of C++ dropped marginally in recent years but is not significant enough to cause alarm. A very popular open source compiler of C++ for the Linux platform is *g++*. Some of the popular open source projects developed with C++ include Mozilla Firefox, MySQL, Android, Blender, LibreCAD, 7-Zip, Mozilla Thunderbird, Chromium, Wireshark, VLC Media Player, VirtualBox, Doxygen, CMake, MariaDB, Code::Blocks, SciPy, PHP and MongoDB. With so much open source software being developed with C++, we definitely can assign it a high Openness Score.

C#

C# is a programming language developed by Microsoft. It is used to develop Web, mobile and enterprise applications. C# supports imperative, functional and object-oriented programming paradigms. Though there is some ambiguity, C# is considered a compiled language by most programmers. C# is still very popular with a top 5 finish in three of the four rankings. Its popularity has declined a bit in the past few years. But we still need to wait further to predict a clear trend regarding its waning popularity. Mono is an open source implementation of Microsoft's .NET Framework



Figure 2: Mono logo

based on the ECMA standards for C# and the Common Language Runtime. It is developed by Xamarin, a subsidiary of Microsoft. Due to the existence of the open source compiler Mono, C# can also be deemed as an open source programming language. Figure 2 shows the logo of Mono. C# is not heavily used in open source software development, featuring only in relatively minor projects like EasyERP, MixERP, uManage, ColorSharp, etc. So, we can assign a low Openness Score to C#.

Go

Go is a relatively new programming language that made its first appearance in 2009. It is a compiled programming language created by Google. It is mainly used for the development of enterprise and Web applications. It supports concurrent and imperative programming paradigms. There are two open source compilers for Go, *gc* from Google and *gccgo* from *GCC*. *gc* is a cross-platform compiler and that makes Go a true open source programming language. The popularity of Go is increasing in all four of the rankings. If you really want to be a part of open source software development, then Go is the next language you have to learn. Docker is a popular open source software developed using Go. The file hosting service Dropbox, with free and proprietary components, is also developed using Go. So, we can safely assign it a medium Openness Score.

PHP

PHP is a general-purpose scripting language suitable for server-side Web development. It is an interpreted language with interpreters for almost all platforms. The existence of *Zend Engine*, an open source interpreter, makes PHP an open source programming language. PHP is still very popular with a Top 3 finish in two of the rankings. The popularity of PHP has also stabilised over the years and there appears no imminent threat to its position. PHP is extensively used in the development of open source projects like Drupal, Joomla, WordPress, Moodle, etc. PHP based open source project development is not nearly as pervasive as C or C++ based development, so we will assign it a medium Openness Score.

Python

Python is a high-level, general-purpose, interpreted, dynamic programming language. It is used for the development of enterprise and Web applications. It supports object-oriented, imperative and functional programming paradigms. An open source interpreter called CPython is the default reference of Python. So, Python can definitely be called an open source programming language. It is still very popular with Top